

# 見 積 書

## 현대위아(주) 황정일과장 貴下

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|--------|----------------|
| 견적일자   | 2017-01-03     |
| 견적번호   | DD20160930-1   |
| 납 기    | 입하 2017년 9월    |
|        | 최종검수 2017년 10월 |
| 견적유효기간 | 1개월            |
| 인도장소   | 멕시코 몬테레이       |
| 지불조건   | 귀사 지불 조건       |

## 대덕테크(주)



대표이사 : 배 태규

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공사명 : 현대위아 멕시코 공장 CVJ POWDER COATING LINE 신설공사

견적금액: 一金 사억육천만원整 ( ₩460,000,000 )

( VAT 별도 )

| NO      | 상세품목내역                  | 수 량    | 단 가        | 금 액                  |
|---------|-------------------------|--------|------------|----------------------|
| 1       | WASHING IN-PUT C/V      | 1 LOT  | 5,000,000  | 5,000,000            |
| 2       | IN-PUT 2 AXLE LOADER    | 1 LOT  | 13,000,000 | 13,000,000           |
| 3       | WASHING BOOTH           | 1 LOT  | 52,000,000 | 52,000,000           |
| 4       | SPRAY BOOTH             | 2 LOT  | 16,500,000 | 33,000,000           |
| 5       | ROBOT                   | 2 LOT  | 9,550,000  | 19,100,000           |
| 6       | SAFETY FENCE            | 1 LOT  | 4,700,000  | 4,700,000            |
| 7       | OVEN IN-PUT 정렬 C/V      | 1 LOT  | 4,700,000  | 4,700,000            |
| 8       | OVEN IN-PUT PUSHER UNIT | 1 LOT  | 3,900,000  | 3,900,000            |
| 9       | OVEN                    | 1 LOT  | 55,500,000 | 55,500,000           |
| 10      | COOLING UNIT            | 1 LOT  | 1,400,000  | 1,400,000            |
| 11      | OVEN OUT-PUT PULL UNIT  | 1 LOT  | 6,000,000  | 6,000,000            |
| 12      | OVEN OUT-PUT 정렬 C/V     | 1 LOT  | 4,900,000  | 4,900,000            |
| 13      | OUT-PUT 3 AXLE LOADER   | 1 LOT  | 18,300,000 | 18,300,000           |
| 14      | STOCK CONVEYOR          | 1 LOT  | 9,000,000  | 9,000,000            |
| 15      | 전기(세척, 건조로)             | 2 SETS | 80,000,000 | 80,000,000           |
| 16      | OFF LINE TEST           | 1 LOT  | 15,000,000 | 15,000,000           |
| 17      | 포장 & F.O.B              | 5 C/T  | 12,000,000 | 12,000,000           |
| 18      | 현지설치작업                  | 1 LOT  | 31,250,000 | 31,250,000           |
| 19      | 부대비용( 항공권&체제비)          | 1 LOT  | 39,750,000 | 39,750,000           |
| 20      | SPARE PARTS             | 1 LOT  | 10,000,000 | 10,000,000           |
| 직접비 합 계 |                         |        |            | 418,500,000          |
| 20      | 일반관리비 & 기업이윤            | 10%    |            | 41,500,000           |
| 견적 합계   |                         |        |            | <b>₩ 460,000,000</b> |

공개입찰가 (견적時 공란)

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1. VAT 제외
2. 1차 UTILITY 공사 제외 ( 전기, 에어, 용수 )
3. ROBOT BODY & CONTROLLER 사급 ( ABB ), GRIPPER & PLC 제어반은 견적포함
4. 약품 및 MASKING JIG SPARE 제외
5. 기타 견적사양외 내용 제외

| DESCRIPTION                                            | SPECIFICATION        | MAT'L  | Q'TY | UNIT | UNIT PRICE | AMOUNT            | REMARK   |
|--------------------------------------------------------|----------------------|--------|------|------|------------|-------------------|----------|
| <b>1. WASHING LINE</b>                                 |                      |        |      |      |            | <b>70,000,000</b> |          |
| <b>1-1. IN-PUT 정렬 CONVEYOR ( W165×L1400×H950 )</b>     |                      |        |      |      |            |                   |          |
| 1) CONVEYOR BODY                                       | W165×L1400×H950      | SUS304 | 1    | LOT  | 600,000    | 600,000           |          |
| 2) MOTOR                                               | 5RK90GU-AWMU+5GU180K | SUS304 | 1    | SET  | 400,000    | 400,000           | ORIENTAL |
| 3) TT CHAIN                                            | TT1143S              | SUS304 | 3    | M    | 70,000     | 210,000           | 한국체인     |
| 4) TT CHAIN SPROCKET                                   | TT 60                | SUS304 | 4    | EA   | 130,000    | 520,000           | 가공       |
| 5) DRIVING SHAFT                                       | Φ30 × 260L           | S45C   | 1    | EA   | 70,000     | 70,000            | 가공       |
| 6) IDLE SHAFT                                          | Φ25 × 230L           | S45C   | 1    | EA   | 70,000     | 70,000            | 가공       |
| 7) IDLE ROLLER                                         | Φ25 × 160L           | S45C   | 2    | EA   | 60,000     | 120,000           | 가공       |
| 8) DRIVING SPROCKET                                    | RS40×B               | S45C   | 2    | EA   | 25,000     | 50,000            | 가공       |
| 9) MC ROLLER                                           | Φ70 × 140L × B/L     | MC     | 2    | EA   | 65,000     | 130,000           | 가공       |
| 10) MC ROLLER                                          | Φ70 × 60L × B/G      | MC     | 1    | EA   | 70,000     | 70,000            | 가공       |
| 11) LASER & BENDING                                    |                      |        | 1    | LOT  | 130,000    | 130,000           | 가공       |
| 12) 잡자재 및 소모품                                          | B/G 외                |        | 1    | LOT  | 180,000    | 180,000           |          |
| 13) 제작인건비                                              | 3인×5일                |        | 15   | M/D  | 150,000    | 2,250,000         |          |
| 14) PAINTING                                           | 분체도장                 |        | 1    | LOT  | 200,000    | 200,000           |          |
| <b>SUB TOTAL</b>                                       |                      |        |      |      |            | <b>5,000,000</b>  |          |
| <b>1-2. IN-PUT 2 AXLE LOADER ( W410×L1340×H2,000 )</b> |                      |        |      |      |            |                   |          |
| 1) LOADER FRAME                                        | SQ 200×150           | SS400  | 1    | LOT  | 900,000    | 900,000           |          |
| 2) X AXLE DRIVE UNIT                                   | MY3M40-800           |        | 1    | SET  | 1,000,000  | 1,000,000         | SMC      |
| 3) Y AXLE DRIVE UNIT                                   | CDAF63-300           |        | 1    | SET  | 600,000    | 600,000           | SMC      |
| 4) AIR CHUCK                                           | 11-MHL2-25D          |        | 2    | SET  | 600,000    | 1,200,000         | SMC      |
| 5) GUIDE TRACK                                         | Φ30 × 800L           | SUJ    | 1    | SET  | 1,200,000  | 1,200,000         | 가공       |
| 6) L/M GUIDE                                           | SL30UU+SR30W         | SUS304 | 1    | SET  | 900,000    | 900,000           | THK      |
| 7) GRIPPER JIG                                         | 1 FINGER             | SUJ    | 1    | SET  | 1,000,000  | 1,000,000         | 가공       |
| 8) GRIPPER UNIT                                        | SOL 4연식              |        | 1    | SET  | 1,000,000  | 1,000,000         | SMC      |
| 9) LASER & BENDING                                     |                      |        | 1    | LOT  | 500,000    | 500,000           |          |
| 10) 잡자재 및 소모품                                          | B/G 외                |        | 1    | LOT  | 550,000    | 550,000           |          |
| 11) 제작인건비                                              | 3인×8일                |        | 25   | M/D  | 150,000    | 3,750,000         |          |
| 12) PAINTING                                           | 분체도장                 |        | 1    | LOT  | 400,000    | 400,000           |          |
| <b>SUB TOTAL</b>                                       |                      |        |      |      |            | <b>13,000,000</b> |          |
| <b>1-3. WASHING BOOTH ( W1,400×L4,200×H1,600 )</b>     |                      |        |      |      |            |                   |          |
| 1) CASING FRAME                                        | SQ 50×50외            | SUS304 | 1    | LOT  | 4,200,000  | 4,200,000         | 제작품      |
| 2) CASING                                              | PLATE 2t             | SUS304 | 1    | LOT  | 2,600,000  | 2,600,000         | 제작품      |
| 3) MOTOR&REDUCER                                       | DWBM80×1/600         |        | 1    | SET  | 1,800,000  | 1,800,000         | 삼양       |
| 4) MESH BELT CHAIN                                     | 1"×1" × ( W 21" )    |        | 12   | M    | 320,000    | 3,840,000         | KEYSTONE |
| 5) SPROCKET                                            | 1"×1" × 23NT         | S45C   | 8    | EA   | 80,000     | 640,000           | 가공       |
| 6) TORQUE LIMIT                                        | TL-10-12             | S45C   | 1    | SET  | 500,000    | 500,000           | 가공       |
| 7) DRIVING SHAFT                                       | Φ40×700L             | S45C   | 1    | SET  | 300,000    | 300,000           | 가공       |
| 8) IDLE SHAFT                                          | Φ40×600L             | S45C   | 1    | SET  | 250,000    | 250,000           | 가공       |
| 9) TAKE UP                                             | Φ50×600L + TENSION   | S45C   | 1    | SET  | 250,000    | 250,000           | 가공       |
| 10) DEGREASING TANK                                    | PLATE 3t             | SUS304 | 1    | LOT  | 1,800,000  | 1,800,000         | 제작품      |
| 11) HOT RINSE TANK                                     | PLATE 3t             | SUS304 | 1    | LOT  | 1,000,000  | 1,000,000         | 제작품      |
| 12) TANK 보온                                            | GLASSWOOL 50t        | G/W    | 1    | LOT  | 300,000    | 300,000           | 벽산       |
| 13) 보온재 COVER                                          | PLATE 1.6t           | EGI    | 1    | LOT  | 600,000    | 600,000           | 제작품      |

| DESCRIPTION         | SPECIFICATION         | MAT'L  | Q'TY | UNIT | UNIT PRICE | AMOUNT            | REMARK       |
|---------------------|-----------------------|--------|------|------|------------|-------------------|--------------|
| 14) OVER FLOW       | PLATE 3t + 40A        | SUS304 | 1    | LOT  | 300,000    | 300,000           | 제작품          |
| 15) DRAIN & PUMP    | 50A                   | SUS304 | 1    | LOT  | 800,000    | 800,000           | 제작품          |
| 16) MESH FILTER     | #40 300×300×20t       | SUS304 | 1    | SET  | 400,000    | 400,000           | 제작품          |
| 17) SPRAY PIPING    | 25A + 보온              | SUS304 | 1    | LOT  | 800,000    | 800,000           | 제작품          |
| 18) SPRAY NOZZLE    | FULL CONE 1/4" 8ℓ/min | SUS304 | 65   | EA   | 10,000     | 650,000           | SPRAY SYSTEM |
| 19) HEATER          | TUBE 10kw - 1,000mmL  | SUS304 | 6    | EA   | 40,000     | 240,000           | 동우히터         |
| 20) 탈지 순환 PUMP      | MTR8-2/2 × 1.1kw      |        | 1    | SET  | 1,500,000  | 1,500,000         | GRUNDFOS     |
| 20) 탈지 SPRAY PUMP   | MTR15-2/2× 3kw        |        | 1    | SET  | 2,500,000  | 2,500,000         | GRUNDFOS     |
| 21) OIL SKIMER      | BELT TYPE / S8125GS-T | SUS304 | 1    | SET  | 1,600,000  | 1,600,000         | 제작품          |
| 22) 탕세 SPRAY PUMP   | MTR8-2/2 × 1.1kw      |        | 1    | SET  | 1,300,000  | 1,300,000         | GRUNDFOS     |
| 23) AIR BLOW NOZZLE | 193ℓ/min              | P.P    | 40   | EA   | 8,000      | 320,000           | SPRAY SYSTEM |
| 24) SOL VALVE       | 25A                   | SUS304 | 4    | SET  | 140,000    | 560,000           |              |
| 25) DRY HEATER      | 10,000 "S" 13kw       |        | 1    | SET  | 1,400,000  | 1,400,000         | LEISTER      |
| 26) RING BLOWER     | 629 3.7kw             |        | 1    | SET  | 1,000,000  | 1,000,000         | HAESUNG      |
| 27) MIST CLEALER    | UCK5-DLH              | SUS304 | 1    | LOT  | 3,000,000  | 3,000,000         | REVEN        |
| 28) EXHAUST DUCT    | 125A~200A PIPE        | SUS304 | 1    | LOT  | 1,300,000  | 1,300,000         | 제작품          |
| 29) DAMPER          | 125A                  | SUS304 | 4    | EA   | 100,000    | 400,000           | 제작품          |
| 30) LASER & BENDING |                       |        | 1    | LOT  | 1,600,000  | 1,600,000         | 가공           |
| 31) 잡자재 및 소모품       | UCP B/G, FITTING류 외   |        | 1    | LOT  | 1,000,000  | 1,000,000         |              |
| 32) 제작인건비           | 3인×25일                |        | 75   | M/D  | 150,000    | 11,250,000        |              |
| 33) PAINTING        | 분체도장                  |        | 1    | LOT  | 2,000,000  | 2,000,000         |              |
| <b>SUB TOTAL</b>    |                       |        |      |      |            | <b>52,000,000</b> |              |

## 2. POWER COATING BOOTH LINE

**56,800,000**

### 2-1. SPRAY BOOTH ( W1,150×L850×H2,100 ) - 2 SETS

|                      |                     |        |   |     |           |            |          |
|----------------------|---------------------|--------|---|-----|-----------|------------|----------|
| 1) FRAME & CASING    | W1,150×L850×H2,100  | SUS304 | 2 | LOT | 1,300,000 | 2,600,000  | 제작품      |
| 2) WALKWAY           | W1,000×L1,700       | SS400  | 1 | SET | 800,000   | 800,000    | 제작품      |
| 3) TURNNING MOTOR    | 5RK90GU-AMU+50GU15K |        | 4 | SET | 250,000   | 1,000,000  | ORIENTAL |
| 4) TURNNING BASE     | W500×L300×8t        | SS400  | 2 | EA  | 100,000   | 200,000    | 제작품      |
| 5) TURNNING HOUSING  | Φ130×L180           | SS400  | 4 | EA  | 160,000   | 640,000    | 제작품      |
| 6) TURNNING HOLDER   |                     | SS400  | 4 | EA  | 200,000   | 800,000    | 제작품      |
| 7) TURNNING FIX NUT  |                     | SS400  | 4 | EA  | 50,000    | 200,000    | 제작품      |
| 8) COUPLING          | AL-SDWA-39C         |        | 4 | EA  | 40,000    | 160,000    | KS       |
| 9) MASKING PLATE     | W120×L330×10t       | SUJ2   | 2 | EA  | 95,000    | 190,000    | 제작품      |
| 10) MASKING CYLINDER | SMC - 63 - 300ST    |        | 2 | EA  | 180,000   | 360,000    | SMC      |
| 11) MASKING BODY     | Φ135×L125           | M.C    | 4 | EA  | 60,000    | 240,000    | 가공       |
| 12) MASKING CAP      | Φ135×L30            | M.C    | 4 | EA  | 50,000    | 200,000    | 가공       |
| 13) MASKING JOINT    | Φ112×L75            | M.C    | 4 | EA  | 60,000    | 240,000    | 가공       |
| 14) MASKING BUSHING  | Φ75×L16             | M.C    | 8 | EA  | 50,000    | 400,000    | 가공       |
| 15) GUIDE BAR        | Φ25×L610            | SUJ2   | 4 | EA  | 60,000    | 240,000    | 가공       |
| 16) EXHAUST FAN      | 27CMM×205mmAq×0.6kw |        | 2 | EA  | 300,000   | 600,000    | 제작품      |
| 17) CARTRIDGE FILTER | Φ270×L600           |        | 4 | EA  | 50,000    | 200,000    | 조일기계     |
| 18) PULSE NOZZLE     | 20A                 |        | 4 | EA  | 60,000    | 240,000    | 조일기계     |
| 19) PULSE VALVE      | 20A                 |        | 4 | EA  | 60,000    | 240,000    | 조일기계     |
| 20) POWERDER GUN     | PEA-C3              |        | 4 | SET | 3,000,000 | 12,000,000 | WAGNER   |
| 21) AIR UNIT         | SOL 5연식             |        | 2 | SET | 1,000,000 | 2,000,000  | SMC      |

| DESCRIPTION                                                 | SPECIFICATION           | MAT'L  | Q'TY | UNIT | UNIT PRICE | AMOUNT      | REMARK   |
|-------------------------------------------------------------|-------------------------|--------|------|------|------------|-------------|----------|
| 22) LASER & BENDING                                         |                         |        | 2    | LOT  | 500,000    | 1,000,000   | 가공       |
| 23) 잡자재 및 소모품                                               | B/G 외                   |        | 1    | LOT  | 300,000    | 300,000     |          |
| 24) 제작인건비                                                   | 3인×15일                  |        | 45   | M/D  | 150,000    | 6,750,000   |          |
| 25) PAINTING                                                | 분체도장                    |        | 2    | LOT  | 700,000    | 1,400,000   |          |
| SUB TOTAL                                                   |                         |        |      |      |            | 33,000,000  |          |
| <b>2-2. ROBOT - 2 SETS</b>                                  |                         |        |      |      |            |             |          |
| 1) ROBOT BODY                                               | IRB 4600-60/IRC5-AXIS 6 |        | 2    | SET  | 위아사급       |             | ABB      |
| 2) ROBOT CONTROLLER                                         | IRC 5                   |        | 2    | SET  | 위아사급       |             | ABB      |
| 3) ROBOT GRIPPER                                            | 2 FINGER CLAMP          |        | 2    | SET  | 2,000,000  | 4,000,000   | 제작품      |
|                                                             | 파지 GRIPPER              |        | 2    | SET  | 500,000    | 1,000,000   |          |
| 4) SOL BLOCK                                                | D-NET 통신                |        | 2    | SET  | 1,000,000  | 2,000,000   | SMC      |
| 5) HOSE , CABLE BRKT                                        |                         | AL     | 2    | SET  | 300,000    | 600,000     | 제작품      |
| 6) AIR HOSE & FITTING                                       |                         | SUS304 | 1    | LOT  | 1,000,000  | 1,000,000   | 구매품      |
| 7) ROBOT 조립                                                 | 3인×10일                  |        | 30   | M/D  | 150,000    | 4,500,000   |          |
| 8) ROBOT TEACHING                                           | 3인×13일                  |        | 40   | M/D  | 150,000    | 6,000,000   |          |
| SUB TOTAL                                                   |                         |        |      |      |            | 19,100,000  |          |
| <b>2-3. SAFETY FENCE</b>                                    |                         |        |      |      |            |             |          |
| 1) PARTITION                                                | 1,000W × 2,000H         |        | 15   | EA   | 250,000    | 3,750,000   | 제작품      |
| 2) DOOR                                                     | 1,000W × 2,000H         |        | 2    | EA   | 300,000    | 600,000     | 제작품      |
| 3) SAFETY CONNECTOR                                         |                         |        | 2    | SET  | 100,000    | 200,000     |          |
| 4) 잡자재 및 소모품                                                |                         |        | 1    | LOT  | 150,000    | 150,000     |          |
| SUB TOTAL                                                   |                         |        |      |      |            | 4,700,000   |          |
| <b>3. OVEN LINE</b>                                         |                         |        |      |      |            | 103,700,000 |          |
| <b>3-1. IN-PUT 정렬 CONVEYOR ( W165×L1880×H950 )</b>          |                         |        |      |      |            |             |          |
| 1) CONVEYOR BODY                                            | W165×L1400×H950         | SUS304 | 1    | LOT  | 500,000    | 500,000     |          |
| 2) MOTOR                                                    | 5RK90GU-AWMU+5GU180K    | SUS304 | 1    | SET  | 300,000    | 300,000     | ORIENTAL |
| 3) TT CHAIN                                                 | TT1143S                 | SUS304 | 4    | M    | 40,000     | 160,000     | 한국체인     |
| 4) TT CHAIN SPROCKET                                        | TT 60                   | SUS304 | 4    | EA   | 120,000    | 480,000     | 가공       |
| 5) DRIVING SHAFT                                            | Φ30 × 260L              | S45C   | 1    | EA   | 50,000     | 50,000      | 가공       |
| 6) IDLE SHAFT                                               | Φ25 × 230L              | S45C   | 1    | EA   | 50,000     | 50,000      | 가공       |
| 7) IDLE ROLLER                                              | Φ25 × 160L              | S45C   | 2    | EA   | 50,000     | 100,000     | 가공       |
| 8) DRIVING SPROCKET                                         | RS40×B                  | S45C   | 2    | EA   | 20,000     | 40,000      | 가공       |
| 9) MC ROLLER                                                | Φ70 × 140L × B/L        | MC     | 2    | EA   | 60,000     | 120,000     | 가공       |
| 10) MC ROLLER                                               | Φ70 × 60L × B/G         | MC     | 1    | EA   | 40,000     | 40,000      | 가공       |
| 11) LASER & BENDING                                         |                         |        | 1    | LOT  | 210,000    | 210,000     | 가공       |
| 12) 잡자재 및 소모품                                               | B/G 외                   |        | 1    | LOT  | 200,000    | 200,000     |          |
| 13) 제작인건비                                                   | 3인×5일                   |        | 15   | M/D  | 150,000    | 2,250,000   |          |
| 14) PAINTING                                                | 분체도장                    |        | 1    | LOT  | 200,000    | 200,000     |          |
| SUB TOTAL                                                   |                         |        |      |      |            | 4,700,000   |          |
| <b>3-2. IN-PUT PUSHER UNIT ( W,1000×L485×H950 - 350ST )</b> |                         |        |      |      |            |             |          |
| 1) FRAME & LEG                                              | W,1000×L485×H950        | SS400  | 1    | LOT  | 300,000    | 300,000     |          |
| 2) AIR CYLINDER                                             | MY1CW32-350ST           |        | 1    | SET  | 800,000    | 800,000     | SMC      |
| 3) HOLDING JIG                                              | W,1000 × 8 CIRCLE       | M.C    | 1    | LOT  | 1,200,000  | 1,200,000   | 가공       |
| 4) LASER & BENDING                                          |                         |        | 1    | LOT  | 130,000    | 130,000     | 가공       |

| DESCRIPTION                                                | SPECIFICATION        | MAT'L  | Q'TY | UNIT | UNIT PRICE | AMOUNT     | REMARK  |
|------------------------------------------------------------|----------------------|--------|------|------|------------|------------|---------|
| 5) 잡자재 및 소모품                                               | B/G 외                |        | 1    | LOT  | 120,000    | 120,000    |         |
| 6) 제작인건비                                                   | 3인×4일                |        | 12   | M/D  | 100,000    | 1,200,000  |         |
| 7) PAINTING                                                | 분체도장                 |        | 1    | LOT  | 150,000    | 150,000    |         |
| SUB TOTAL                                                  |                      |        |      |      |            | 3,900,000  |         |
| <b>3-3. OVEN ( W1,730×L4,400×H1,800 )</b>                  |                      |        |      |      |            |            |         |
| 1) FRAME & LEG                                             | W1,730×L4,400×H1,800 | SS400  | 1    | LOT  | 2,800,000  | 2,800,000  |         |
|                                                            | L65×65×6t            | SS400  | 1    | LOT  | 320,000    | 320,000    |         |
|                                                            | L50×50×6t            | SS400  | 1    | LOT  | 350,000    | 350,000    |         |
| 2) CASING                                                  | PLATE 3t             | SUS304 | 1    | LOT  | 3,000,000  | 3,000,000  |         |
|                                                            | PLATE 1.5t           | EGI    | 1    | LOT  | 2,000,000  | 2,000,000  |         |
| 3) INSULATION                                              | R/W 150t #80         | R/W    | 1    | LOT  | 1,400,000  | 1,400,000  |         |
| 4) INSPECTION DOOR                                         |                      | SUS304 | 3    | SET  | 300,000    | 900,000    |         |
| 5) GEARD MOTOR                                             | DWBM120 × 1/30       |        | 1    | SET  | 500,000    | 500,000    | 삼양      |
| 6) REDUCER                                                 | DWD100 × 1/600       |        | 1    | SET  | 800,000    | 800,000    | 삼양      |
| 7) FLAT WIRE CHAIN                                         | 1"×1" × ( W 53" )    |        | 15   | M    | 800,000    | 12,000,000 | KYSTONE |
| 8) SPROCKET                                                | P25.4 × 23NT - B/R   | S45C   | 18   | EA   | 150,000    | 2,700,000  | 가공      |
| 9) MC ROLLER                                               | Φ172.8 × Φ60 × L200  | M.C    | 6    | EA   | 150,000    | 900,000    | 가공      |
| 10) TORQUE LIMIT                                           | TL-700-2             | S45C   | 1    | EA   | 500,000    | 500,000    | 가공      |
| 11) DRIVE SHAFT                                            | Φ60 × L1,700         | S45C   | 1    | EA   | 250,000    | 250,000    | 가공      |
| 12) IDLE SHAFT                                             | Φ50 × L1,700         | S45C   | 6    | EA   | 180,000    | 1,080,000  | 가공      |
| 13) TUBE HEATER                                            | 1.25kW - 1,400L U    | SUS304 | 15   | EA   | 60,000     | 900,000    | 동우히터    |
| 14) TURBO FAN                                              | 80CMM×80mmAq×5HP     | SUS304 | 1    | SET  | 2,600,000  | 2,600,000  | 제작품     |
| 15) R.C FAN                                                | 30CMM×35mmAq×1.5HP   | SUS304 | 3    | SET  | 1,800,000  | 5,400,000  | 제작품     |
| 16) EXHAUST HOOD                                           | W1,730×H400          | SUS304 | 2    | SET  | 300,000    | 600,000    | 제작품     |
| 17) SPIRAL DUCT                                            | 250A                 |        | 30   | M    | 30,000     | 900,000    | 구매품     |
| 18) LASER & BENDING                                        |                      |        | 1    | LOT  | 1,500,000  | 1,500,000  | 가공      |
| 19) 잡자재 및 소모품                                              | B/G 외                |        | 1    | LOT  | 600,000    | 600,000    |         |
| 20) 제작인건비                                                  | 4인×20일               |        | 70   | M/D  | 150,000    | 10,500,000 |         |
| 21) PAINTING                                               | 분체도장                 |        | 1    | LOT  | 3,000,000  | 3,000,000  |         |
| SUB TOTAL                                                  |                      |        |      |      |            | 55,500,000 |         |
| <b>3-4. COOLING UNIT ( W,1300×L490×H350 )</b>              |                      |        |      |      |            |            |         |
| 1) FRAME & LEG                                             | W,1300×L490×H350     | SUS304 | 1    | LOT  | 250,000    | 250,000    |         |
| 2) COOLING BLOWER                                          | DR-DF25ASB           |        | 3    | SET  | 60,000     | 180,000    | 제작품     |
| 3) LASER & BENDING                                         |                      |        | 1    | LOT  | 60,000     | 60,000     | 가공      |
| 4) 잡자재 및 소모품                                               | B/G 외                |        | 1    | LOT  | 60,000     | 60,000     |         |
| 5) 제작인건비                                                   | 2인×3일                |        | 5    | M/D  | 150,000    | 750,000    |         |
| 6) PAINTING                                                | 분체도장                 |        | 1    | LOT  | 100,000    | 100,000    |         |
| SUB TOTAL                                                  |                      |        |      |      |            | 1,400,000  |         |
| <b>3-5. OUT-PUT PULL UNIT ( W,1120×L920×H950 - 350ST )</b> |                      |        |      |      |            |            |         |
| 1) FRAME & LEG                                             | W,1120×L920×H950     | SS400  | 1    | LOT  | 400,000    | 400,000    |         |
| 2) AIR CYLINDER                                            | MY1CW32-350ST        |        | 1    | SET  | 700,000    | 700,000    | SMC     |
| 3) MAGNETIC                                                | 33 × 80 DC110V       |        | 8    | SET  | 300,000    | 2,400,000  |         |
| 4) LASER & BENDING                                         |                      |        | 1    | LOT  | 250,000    | 250,000    | 가공      |
| 5) 잡자재 및 소모품                                               | B/G 외                |        | 1    | LOT  | 270,000    | 270,000    |         |
| 6) 제작인건비                                                   | 3인×4일                |        | 12   | M/D  | 150,000    | 1,800,000  |         |
| 7) PAINTING                                                | 분체도장                 |        | 1    | LOT  | 180,000    | 180,000    |         |

| DESCRIPTION                                                | SPECIFICATION        | MAT'L  | Q'TY | UNIT | UNIT PRICE | AMOUNT     | REMARK   |
|------------------------------------------------------------|----------------------|--------|------|------|------------|------------|----------|
| SUB TOTAL                                                  |                      |        |      |      |            | 6,000,000  |          |
| <b>3-6. OVEN OUT-PUT 정렬 CONVEYOR ( W165×L1980×H950 )</b>   |                      |        |      |      |            |            |          |
| 1) CONVEYOR BODY                                           | W165×L1400×H950      | SUS304 | 1    | LOT  | 500,000    | 500,000    |          |
| 2) MOTOR                                                   | 5RK90GU-AWMU+5GU180K | SUS304 | 1    | SET  | 400,000    | 400,000    | ORIENTAL |
| 3) TT CHAIN                                                | TT1143S              | SUS304 | 4    | M    | 40,000     | 160,000    | 한국체인     |
| 4) TT CHAIN SPROCKET                                       | TT 60                | SUS304 | 4    | EA   | 140,000    | 560,000    | 가공       |
| 5) DRIVING SHAFT                                           | Φ30 × 260L           | S45C   | 1    | EA   | 60,000     | 60,000     | 가공       |
| 6) IDLE SHAFT                                              | Φ25 × 230L           | S45C   | 1    | EA   | 60,000     | 60,000     | 가공       |
| 7) IDLE ROLLER                                             | Φ25 × 160L           | S45C   | 2    | EA   | 50,000     | 100,000    | 가공       |
| 8) DRIVING SPROCKET                                        | RS40×B               | S45C   | 2    | EA   | 15,000     | 30,000     | 가공       |
| 9) MC ROLLER                                               | Φ70 × 140L × B/L     | MC     | 2    | EA   | 60,000     | 120,000    | 가공       |
| 10) MC ROLLER                                              | Φ70 × 60L × B/G      | MC     | 1    | EA   | 40,000     | 40,000     | 가공       |
| 11) LASER & BENDING                                        |                      |        | 1    | LOT  | 170,000    | 170,000    | 가공       |
| 12) 잡자재 및 소모품                                              | B/G 외                |        | 1    | LOT  | 250,000    | 250,000    |          |
| 13) 제작인건비                                                  | 3인×5일                |        | 15   | M/D  | 150,000    | 2,250,000  |          |
| 14) PAINTING                                               | 분체도장                 |        | 1    | LOT  | 200,000    | 200,000    |          |
| SUB TOTAL                                                  |                      |        |      |      |            | 4,900,000  |          |
| <b>3-7. OUT-PUT 3 AXLE LOADER ( W1,100×L1,400×H2,000 )</b> |                      |        |      |      |            |            |          |
| 1) LOADER FRAME                                            | SQ 200×150           | SS400  | 1    | LOT  | 1,200,000  | 1,200,000  |          |
| 2) X AXLE DRIVE UNIT                                       | MY3M40-1000          |        | 1    | SET  | 1,500,000  | 1,500,000  | SMC      |
| 3) Y AXLE DRIVE UNIT                                       | MY3M40-800           |        | 1    | SET  | 1,400,000  | 1,400,000  | SMC      |
| 4) Z AXLE DRIVE UNIT                                       | CDAF63-300           |        | 1    | SET  | 400,000    | 400,000    | SMC      |
| 5) AIR CHUCK                                               | 11-MHL2-25D          |        | 4    | SET  | 500,000    | 2,000,000  | SMC      |
| 6) GUIDE TRACK                                             | Φ30 × 1,000L,800L    | SUJ    | 2    | SET  | 1,000,000  | 2,000,000  | 가공       |
| 7) L/M GUIDE                                               | SL30UU+SR30W         | SUS304 | 2    | SET  | 800,000    | 1,600,000  | THK      |
| 8) GRIPPER JIG                                             | 2 FINGER             | SUJ    | 1    | SET  | 1,800,000  | 1,800,000  | 가공       |
| 8) GRIPPER UNIT                                            | SOL 4연식              |        | 1    | SET  | 800,000    | 800,000    | SMC      |
| 9) LASER & BENDING                                         |                      |        | 1    | LOT  | 400,000    | 400,000    |          |
| 10) 잡자재 및 소모품                                              | B/G 외                |        | 1    | LOT  | 400,000    | 400,000    |          |
| 11) 제작인건비                                                  | 3인×10일               |        | 30   | M/D  | 150,000    | 4,500,000  |          |
| 12) PAINTING                                               | 분체도장                 |        | 1    | LOT  | 300,000    | 300,000    |          |
| SUB TOTAL                                                  |                      |        |      |      |            | 18,300,000 |          |
| <b>3-8. STOCK CONVEYOR ( W710×L1800×H950 )</b>             |                      |        |      |      |            |            |          |
| 1) FRAME & LEG                                             | W710×L1800×H950      | SS400  | 1    | LOT  | 1,000,000  | 1,000,000  |          |
| 2) MOTOR                                                   | DWBM80×1/600         |        | 1    | SET  | 700,000    | 700,000    |          |
| 3) TT CHAIN                                                | TT1143S×6열           | SUS304 | 25   | M    | 50,000     | 1,250,000  | 한국체인     |
| 4) TT CHAIN SPROCKET                                       | TT 60                | SUS304 | 12   | EA   | 140,000    | 1,680,000  | 가공       |
| 5) DRIVING SHAFT                                           | Φ50 × 800L           | S45C   | 1    | EA   | 60,000     | 60,000     | 가공       |
| 6) IDLE SHAFT                                              | Φ40 × 700L           | S45C   | 1    | EA   | 60,000     | 60,000     | 가공       |
| 7) IDLE ROLLER                                             | Φ25 × 160L           | S45C   | 24   | EA   | 40,000     | 960,000    | 가공       |
| 8) DRIVING SPROCKET                                        | RS40×B               | S45C   | 2    | EA   | 20,000     | 40,000     | 가공       |
| 9) MC ROLLER                                               | Φ70 × 140L × B/L     | MC     | 6    | EA   | 50,000     | 300,000    | 가공       |
| 10) MC ROLLER                                              | Φ70 × 60L × B/G      | MC     | 10   | EA   | 40,000     | 400,000    | 가공       |
| 11) LASER & BENDING                                        |                      |        | 1    | LOT  | 250,000    | 250,000    | 가공       |
| 12) 잡자재 및 소모품                                              | B/G 외                |        | 1    | LOT  | 400,000    | 400,000    |          |
| 13) 제작인건비                                                  | 3인×5일                |        | 10   | M/D  | 150,000    | 1,500,000  |          |

| DESCRIPTION  | SPECIFICATION | MAT'L | Q'TY | UNIT | UNIT PRICE | AMOUNT      | REMARK |
|--------------|---------------|-------|------|------|------------|-------------|--------|
| 14) PAINTING | 분체도장          |       | 1    | LOT  | 400,000    | 400,000     |        |
| SUB TOTAL    |               |       |      |      |            | 9,000,000   |        |
| 기계장치 합계      |               |       |      |      |            | 230,500,000 |        |

#### 4. 전기

##### 4-1. WASHING CONTROL PANEL

|                    |                             |        |           |            |            |
|--------------------|-----------------------------|--------|-----------|------------|------------|
| 1) MCCB            | 3VL4725-1DC36-0AA0(20~150A) | 21 EA  | 200,000   | 4,200,000  | SIEMENS    |
|                    | 5SY5210-7, 202-7            | 8 EA   | 50,000    | 400,000    | SIEMENS    |
| 2) M.G S/W         | 3RT2024,26,28-1AP04         | 24 EA  | 25,000    | 600,000    | SIEMENS    |
| 3) EOCR            | 3BZ2                        | 23 EA  | 100,000   | 2,300,000  | SCHNEIDER  |
| 4) CT              | 200/5VA                     | 4 EA   | 10,000    | 40,000     |            |
| 5) TRANSFORMER     | 480/220/110V 5KVA           | 1 EA   | 700,000   | 700,000    | SIEMENS    |
| 6) TIC             | E5CC-C03ASM-002             | 3 EA   | 70,000    | 210,000    | OMRON      |
| 7) PLC             | SIEMENS S-7                 | 2 EA   | 5,000,000 | 10,000,000 | SIEMENS    |
| 8) I/O BOARD       | R32ST                       | 1 EA   | 250,000   | 250,000    |            |
| 9) NOISER FILTER   | 10A                         | 1 EA   | 30,000    | 30,000     | MURR       |
| 10) POWER SUPPLY   | DC 24V 10A                  | 1 EA   | 300,000   | 300,000    | OMRON      |
| 11) I/O CABLE      | 2M                          | 10 EA  | 25,000    | 250,000    |            |
| 12) TOUCH SCREEN   | 6AV2124-OMC01-OAX0          | 2 EA   | 1,800,000 | 3,600,000  | SIEMENS    |
| 13) PT             | 480/110V                    | 1 EA   | 80,000    | 80,000     |            |
| 14) RELAY          | MY-4N 110V                  | 30 EA  | 5,000     | 150,000    | HONEYWELL  |
| 15) MULTI METER    | KDA-A                       | 1 EA   | 300,000   | 300,000    | LIGHT STAR |
| 16) INVERTER       | 6SE6420-2UD21-5AA1          | 3 EA   | 700,000   | 2,100,000  | SIEMENS    |
| 17) A-METER        | KAA-08 400A 300%            | 3 EA   | 20,000    | 60,000     |            |
| 18) WH-METER       | OMNI-3315                   | 1 EA   | 180,000   | 180,000    |            |
| 19) METER          | MP5W                        | 1 EA   | 60,000    | 60,000     |            |
| 20) SURGE FILTER   | BJLPA3P-LCD DISPLAY TYPE    | 1 EA   | 2,000,000 | 2,000,000  | BJ TECH    |
| 21) SPARKER KILLER | K3CR-50500                  | 26 EA  | 5,000     | 130,000    | KEM        |
| 22) PATROL LAMP    | ST56LBZ32 4RYG SZ24         | 2 EA   | 80,000    | 160,000    | Q-LIGHT    |
| 23) SWITCH         | KH-506                      | 1 LOT  | 400,000   | 400,000    | KOINO      |
| 24) PANEL AIRCON   | AP-ES140                    | 1 EA   | 1,800,000 | 1,800,000  | AIR MAJOR  |
| 25) 조립용 자재         |                             | 1 LOT  | 700,000   | 700,000    |            |
| 26) MAIN PANEL외함   | 1800*2000*500               | 1 EA   | 1,500,000 | 1,500,000  | RITTAL형    |
| 27) LOCAL BOX      | 400*650                     | 2 EA   | 300,000   | 600,000    | RITTAL형    |
| 28) 제작, 조립 인건비     | 3인×10일                      | 30 M/D | 150,000   | 4,500,000  |            |
| SUB TOTAL          |                             |        |           | 37,600,000 |            |

##### 4-2. OVEN CONTROL PANEL

|                |                             |    |    |           |           |           |
|----------------|-----------------------------|----|----|-----------|-----------|-----------|
| 1) MCCB        | 3VL4725-1DC36-0AA0(20~150A) | 14 | EA | 300,000   | 4,200,000 | SIEMENS   |
|                | 5SY5210-7, 202-7            | 7  | EA | 50,000    | 350,000   | SIEMENS   |
| 2) M.G S/W     | 3RT2024,26,28-1AP04         | 10 | EA | 30,000    | 300,000   | SIEMENS   |
| 3) EOCR        | 3BZ2                        | 9  | EA | 120,000   | 1,080,000 | SCHNEIDER |
| 4) CT          | 200/5VA                     | 7  | EA | 10,000    | 70,000    |           |
| 5) TRANSFORMER | 480/220/110V 5KVA           | 1  | EA | 700,000   | 700,000   | SIEMENS   |
| 6) TIC         | E5CC-C03ASM-002             | 3  | EA | 70,000    | 210,000   | OMRON     |
| 7) PLC         | SIEMENS S-7                 | 1  | EA | 4,500,000 | 4,500,000 | SIEMENS   |
| 8) I/O BOARD   | R32ST                       | 1  | EA | 250,000   | 250,000   |           |

| DESCRIPTION          | SPECIFICATION            | MAT'L | Q'TY | UNIT | UNIT PRICE | AMOUNT     | REMARK     |
|----------------------|--------------------------|-------|------|------|------------|------------|------------|
| 9) NOISER FILTER     | 10A                      |       | 1    | EA   | 30,000     | 30,000     | MURR       |
| 10) POWER SUPPLY     | DC 24V 10A               |       | 1    | EA   | 300,000    | 300,000    | OMRON      |
| 11) I/O CABLE        | 2M                       |       | 10   | EA   | 25,000     | 250,000    |            |
| 12) TOUCH SCREEN     | 6AV2124-OMC01-OAX0       |       | 1    | EA   | 1,800,000  | 1,800,000  | SIEMENS    |
| 13) SCR UNIT         | 480V 35A                 |       | 2    | EA   | 500,000    | 1,000,000  | HAN YOUNG  |
| 14) PT               | 480/110V                 |       | 1    | EA   | 80,000     | 80,000     |            |
| 15) RELAY            | MY-4N 110V               |       | 20   | EA   | 5,000      | 100,000    | HONEYWELL  |
| 16) MULTI METER      | KDA-A                    |       | 1    | EA   | 250,000    | 250,000    | LIGHT STAR |
| 17) INVERTER         | 6SE6420-2UD21-5AA1       |       | 3    | EA   | 700,000    | 2,100,000  | SIEMENS    |
| 18) A-METER          | KAA-08 400A 300%         |       | 3    | EA   | 20,000     | 60,000     |            |
| 19) WH-METER         | OMNI-3315                |       | 1    | EA   | 180,000    | 180,000    |            |
| 20) METER            | MP5W                     |       | 1    | EA   | 60,000     | 60,000     |            |
| 21) SURGE FILTER     | BJLPA3P-LCD DISPLAY TYPE |       | 1    | EA   | 2,500,000  | 2,500,000  | BJ TECH    |
| 22) SPARKER KILLER   | K3CR-50500               |       | 8    | EA   | 10,000     | 80,000     | KEM        |
| 23) NOISER FILTER    | ZNC2210-00U              |       | 5    | EA   | 30,000     | 150,000    | TDK        |
| 24) PATROL LAMP      | ST56LBZ32 4RYG SZ24      |       | 2    | EA   | 80,000     | 160,000    | Q-LIGHT    |
| 25) SWITCH           | KH-506                   |       | 1    | LOT  | 400,000    | 400,000    | KOINO      |
| 26) PANEL AIRCON     | AP-ES140                 |       | 1    | EA   | 1,800,000  | 1,800,000  | AIR MAJOR  |
| 27) 조립용 자재           |                          |       | 1    | LOT  | 600,000    | 600,000    |            |
| 28) MAIN PANEL외함     | 1800*2000*500            |       | 1    | EA   | 1,500,000  | 1,500,000  | RITTAL형    |
| 29) LOCAL BOX        | 400*650                  |       | 1    | EA   | 300,000    | 300,000    | RITTAL형    |
| 30) 제작, 조립 인건비       | 3인×8일                    |       | 24   | M/D  | 150,000    | 3,600,000  |            |
| SUB TOTAL            |                          |       |      |      |            | 28,960,000 |            |
| 4-3. 설치자재 및 인건비      |                          |       |      |      |            |            |            |
| 1) CABLE             | LAPP 4G 25SQ 외           |       | 1    | LOT  | 4,000,000  | 4,000,000  | LAPP CABLE |
| 2) HEATING CABLE     | 16SQ                     |       | 1    | LOT  | 500,000    | 500,000    |            |
| 3) CABLE DUCT        | 200*100                  |       | 1    | LOT  | 1,500,000  | 1,500,000  |            |
| 4) SENSOR            | BES Q40KFU-NAC25F-S04G   |       | 3    | EA   | 30,000     | 90,000     | BALLUFF    |
|                      | BES 516-361-E5-Y-S4      |       | 25   | EA   | 30,000     | 750,000    | BALLUFF    |
|                      | BOS 6K-NU-LK10-S75       |       | 11   | EA   | 110,000    | 1,210,000  | BALLUFF    |
|                      | BOS 6K-PU-RD10-S75       |       | 1    | EA   | 150,000    | 150,000    | BALLUFF    |
| 5) SENSOR CABLE      | M12 CONNECTOR 3M         |       | 40   | EA   | 13,000     | 520,000    |            |
| 6) DOOR SAFET SENSOR | MGB-PN                   |       | 3    | EA   | 400,000    | 1,200,000  | EUCHNER    |
| 7) TIC SENSOR        | PT100                    |       | 6    | EA   | 70,000     | 420,000    |            |
| 8) RECODER           | GR-200                   |       | 1    | SET  | 1,400,000  | 1,400,000  |            |
| 9) MAGNETIC CONTROL  | KMS-20                   |       | 1    | SET  | 500,000    | 500,000    |            |
| 10) FLEXIBLE         | 28MM 외                   |       | 1    | LOT  | 400,000    | 400,000    |            |
| 11) 전선관              | 28MM 외                   |       | 1    | LOT  | 300,000    | 300,000    |            |
| 12) 설치잡자재            | CHANNEL 외                |       | 1    | LOT  | 500,000    | 500,000    |            |
| SUB TOTAL            |                          |       |      |      |            | 13,440,000 |            |
| 전기장치 합계              |                          |       |      |      |            | 80,000,000 |            |

## 5. OFF LINE TEST

|           |      |    |     |         |           |
|-----------|------|----|-----|---------|-----------|
| 1) OLT 설치 | 기계 4 | 15 | M/D | 150,000 | 2,250,000 |
|           | 전기 2 | 10 | M/D | 150,000 | 1,500,000 |
| 2) 장비비    |      | 5  | 대   | 300,000 | 1,500,000 |

| DESCRIPTION       | SPECIFICATION   | MAT'L | Q'TY | UNIT | UNIT PRICE | AMOUNT     | REMARK  |
|-------------------|-----------------|-------|------|------|------------|------------|---------|
| 3) PROGRAM & TEST | 기계 2            |       | 15   | M/D  | 150,000    | 2,250,000  |         |
|                   | 전기 2            |       | 20   | M/D  | 150,000    | 3,000,000  |         |
| 4) 해체 작업          | 기계 4            |       | 15   | M/D  | 150,000    | 2,250,000  |         |
|                   | 전기 2            |       | 15   | M/D  | 150,000    | 2,250,000  |         |
| OLT 합계            |                 |       |      |      |            | 15,000,000 |         |
| 6. 포장 & F.O.B     |                 |       |      |      |            |            |         |
| 1) 포장             | WOOD BOX        |       | 4    | C/T  | 2,500,000  | 10,000,000 |         |
| 2) F.O.B 부산       |                 |       | 4    | C/T  | 500,000    | 2,000,000  |         |
| 포장 & F.O.B 합계     |                 |       |      |      |            | 12,000,000 |         |
| 7. 현지 설치작업        |                 |       |      |      |            |            |         |
| 1) 컨테이너 해포 및 운반   | 5 CONTAINER     |       | 5    | M/D  | 250,000    | 1,250,000  | 장비 위아지원 |
| 2) 기계 설치          | 기계 4인×10일       |       | 40   | M/D  | 250,000    | 10,000,000 | 장비 위아지원 |
|                   | 전기 2인×10일       |       | 20   | M/D  | 250,000    | 5,000,000  |         |
| 3) 시운전            | 기계 2인×10일       |       | 20   | M/D  | 250,000    | 5,000,000  |         |
|                   | 전기 1인×10일       |       | 10   | M/D  | 250,000    | 2,500,000  |         |
|                   | ROBOT 티칭 1인×10일 |       | 10   | M/D  | 250,000    | 2,500,000  |         |
| 4) 양산대기           | 기계 1, 전기 1      |       | 20   | M/D  | 250,000    | 5,000,000  |         |
| 현지 설치작업 합계        |                 |       | 125  |      |            | 31,250,000 |         |
| 8. 항공권 및 체제비      |                 |       |      |      |            |            |         |
| 1) 항공권            | 서울 - > 몬테레이     |       | 7    | 회    | 2,600,000  | 18,200,000 | 왕복기준    |
| 2) 현지 체제비         | 숙박 & 식대         |       | 125  | M/D  | 150,000    | 18,750,000 |         |
| 3) DOCUMENTS      |                 |       | 1    | LOT  | 2,800,000  | 2,800,000  |         |
| 항공권 및 체제비 합계      |                 |       |      |      |            | 39,750,000 |         |